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Abstract. Both methods to control the gain in the modulation chain of interferometric fiber optic gyroscope (IFOG) are analyzed: One is based on the square wave biasing modulation, and the other is based on the four-state wave biasing modulation. The analysis results indicate that the method based on square wave is less efficient at low angular rate because of long reset time, and the method based on four-state wave may give the wrong or reverse gain error at high angular rate because of frequent resets. To eliminate the limitation of the existing methods, a novel demodulation method of the gain in the modulation chain based on periodical biasing modulation is proposed. This method is independent of the 2π phase ramp resets and the input angular rate. Experimental results indicate that the method effectively improves the scale factor stability and linearity of closed-loop IFOG. © 2012 Society of Photo-Optical Instrumentation Engineers (SPIE). [DOI: [10.1117/1.OE.51.6.064401](https://doi.org/10.1117/1.OE.51.6.064401)]

Subject terms: periodical biasing modulation; gain error; modulation chain; fiber optic gyroscope.

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1 Introduction

In closed-loop interferometric fiber optic gyroscope (IFOG), the digital serrodyne is usually adopted as the feedback modulation signal to compensate Sagnac phase shift induced by rotation and a bias modulation signal is added simultaneously to guarantee IFOG operating at the maximum sensitivity.^{1,2} The half-wave voltage of the LiNbO₃ based modulator and the reference voltage of D/A convertor both are subjected to temperature fluctuation. As a result, the gain in modulation chain drifts with temperature too. The gain error in modulation chain introduces error into the feedback modulation phase which influences the performance about scale factor of IFOG.^{3,4} To suppress the gain error in modulation chain, the gain is controlled by demodulating the output signals of the detector at and before or after reset when the square wave is adopted as biasing modulation signal.^{5,6} When the four-state wave is adopted as a biasing modulation signal, the gain error can be demodulated according to the output signal at any four adjacent modulation states instead of at the reset.⁷

In actual application, the defects of above both demodulation methods of gain in modulation chain are exposed. At low angular rate, the time interval between two adjacent resets is so long in the former method that the gain cannot be demodulated in a long period, and the control of gain becomes invalid. For the latter method, at high angular rate, the time interval between resets is very short, and the wrong demodulation results are obtained due to frequent resets. To overcome the limitation of the existed gain demodulation methods, a control method based on periodical biasing modulation is supposed in theory and confirmed by experiments here.

2 Influence of Gain Error in Modulation Chain on Scale Factor

The typical controlling circuit in IFOG is shown in Fig. 1. In Fig. 1, $\Delta\phi_R$ is the Sagnac phase shift, $\Delta\phi_F$ is the feedback phase shift, and $\Delta\phi$ is the difference between $\Delta\phi_R$ and $\Delta\phi_F$.

The scale factor of the IFOG controlled by the closed-loop in Fig. 1 is

$$K_{\text{IFOG}} = \frac{D_{\text{out}}(t)}{\Omega} = K_f \cdot \frac{2\pi DL}{\lambda c}, \quad (1)$$

where $D_{\text{out}}(t)$ is the digital output signal at time t of IFOG, Ω is the rotation rate, D is the diameter of the fiber coil and L is the length of fiber, λ is the average wavelength of the source, and c is the optic velocity in vacuum. K_f is the gain in modulation chain. Due to the limitation of D/A convertor and power voltage, it is impossible to realize a continuous serrodyne. In practice, the digital serrodyne with an adequate synchronized reset via automatic overflow of the convertor is adopted, and the overflow value is usually 2π . The gain in modulation chain of IFOG with reset digital serrodyne feedback is⁵

$$K_f = \frac{V_\pi}{\pi} \cdot \frac{2^N}{V_{pp}}, \quad (2)$$

where V_π is the half-wave voltage of the phase modulator, and N is the bit number of D/A convertor. V_{pp} is the output voltage of the D/A convertor at full scale varies with temperature at rate $5 \times 10^{-4}/^\circ\text{C}$, so that the temperature fluctuation of operation environment may cause the instability of the gain K_f in modulation chain. According to Eq. (1), the scale factor of IFOG is proportional to K_f , so the instability of K_f may introduce error in the scale factor.

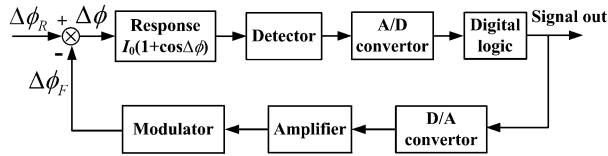


Fig. 1 The schematics of the controlling circuit generally adopted by IFOG.

Equation (1) represents the transient relation between the gain in modulation chain and the scale factor. The scale factor error caused by gain in modulation chain is opposite during the ramp and the reset of the digital serrrodyne so that the measured scale factor is a value consisting of the average error during the digital integral, and the measured scale factor error is proportional to the cubic power of the gain variation in modulation chain.⁸

3 Analysis About Demodulation Methods

3.1 Limitation of Existed Gain Demodulation Methods

The square wave biasing modulation and four-state wave modulation are both popular methods to demodulate the gain error in modulation chain of IFOG. In actual operation, the limitation of both methods is observed.

In the IFOG modulated by square wave, the gain error in modulation chain is demodulated by examining the output signal of detector at and before or after reset, that is

$$\Delta D = D_R - D_{R^+} \quad \text{or} \quad \Delta D = D_R - D_{R^-}, \quad (3)$$

where ΔD is the gain error, D_R , D_{R^+} and D_{R^-} is the sampling value of A/D converter at, before, and after reset, respectively. The gain error can be demodulated only during the reset. In other words, if there is no reset, the gain error cannot be demodulated. For an IFOG with 1000 m fiber in a diameter 80 mm coil, the dynamic range of which is ± 100 deg/s, the reset occurs every other several hundred seconds when the angular rate is 0.01 deg/h. In this situation, correspondingly, the gain error is demodulated once every other several hundred seconds, which is too long to control the gain in the modulation chain; that is, the demodulation is invalid at low angular rate.

In the four-state wave modulation method, the optic signal in IFOG is modulated by the four-state square wave

$$\phi_b(t) = \begin{cases} \phi & 0 \leq t < \tau/2 \\ a\phi & \tau/2 \leq t < \tau \\ -\phi & \tau \leq t < 3\tau/2 \\ -a\phi & 3\tau/2 \leq t < 2\tau \end{cases}, \quad (4)$$

in which ϕ is modulation phase, $\cos(\phi) = \cos(a\phi)$, and τ is the transit time of optic in the fiber coil. If the sampling value of A/D converter is D_x when the modulation phase is x , the gain error in modulation chain is represented as

$$\Delta D = D_\phi + D_{-\phi} - D_{a\phi} - D_{-a\phi}. \quad (5)$$

The mixed modulation signal of the four-state signal and feedback signal imposed on the modulator is illustrated in Fig. 2 in the situation that EMBED Equation. DSMT4,

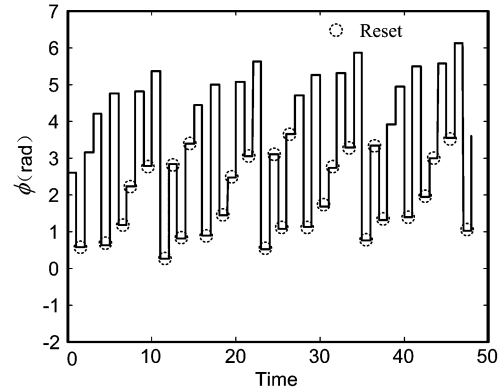


Fig. 2 The supposition of the big feedback signal and four-state biasing modulation signal.

$\phi = \pi/2$, $a = 3$, and the feedback phase is $\pi/3$. From Fig. 2, the signal resets frequently due to the big feedback phase which corresponds to high angular rate. When a simulated relative gain error of 2% is added in modulation chain by operational amplifier, the demodulation signal of gain error is shown in Fig. 3. It is obvious that the gain error cannot be demodulated correctly. If the wrong signal in Fig. 3 is feedback to gain control circuit, the convergent controlling results cannot be obtained; that is, the gain demodulation and controlling method is invalid at high angular rate.

3.2 Gain Demodulation Method Based on Periodic Biasing Modulation

To overcome the limitation of both demodulation methods, a demodulation method based on periodic biasing modulation is supposed. In the method, a four-state biasing signal is inserted periodically in feedback signal. The four-state signal is as

$$\phi_b(t) = \begin{cases} \pi/2 & 0 \leq t < \tau/2 \\ \pi & \tau/2 \leq t < \tau \\ -\pi/2 & \tau \leq t < 3\tau/2 \\ -\pi & 3\tau/2 \leq t < 2\tau \end{cases}. \quad (6)$$

When the biasing modulation signal is inserted, the feedback signal of the IFOG is paused; that is, the feedback signal and the biasing modulation signal have no overlap in time. If the relative phase error in modulation chain is ε , the output signal of the detector in the biasing modulation period is

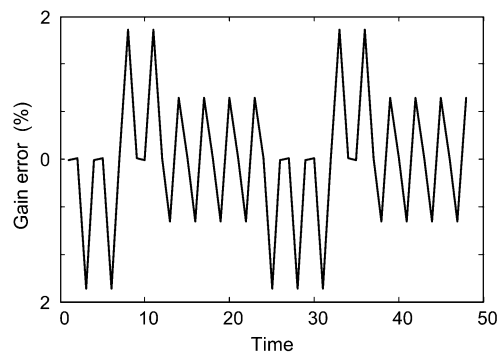


Fig. 3 The gain error demodulated by four-state biasing modulation method.

$$V_1 = K_T I_0 \cos\left[\phi_R + \frac{\pi}{2}(1 + \varepsilon)\right]$$

$$= -K_T I_0 \sin\left(\phi_R + \frac{\pi}{2}\varepsilon\right) \quad 0 \leq t < \tau/2 \quad (7)$$

$$V_2 = K_T I_0 \cos[\phi_R + \pi(1 + \varepsilon)]$$

$$= -K_T I_0 \cos(\phi_R + \pi\varepsilon) \quad \tau/2 \leq t < \tau \quad (8)$$

$$V_3 = K_T I_0 \cos\left[\phi_R - \frac{\pi}{2}(1 + \varepsilon)\right]$$

$$= K_T I_0 \sin\left(\phi_R - \frac{\pi}{2}\varepsilon\right) \quad \tau \leq t < 3\tau/2 \quad (9)$$

$$V_4 = K_T I_0 \cos[\phi_R - \pi(1 + \varepsilon)]$$

$$= -K_T I_0 \cos(\phi_R - \pi\varepsilon) \quad 3\tau/2 \leq t < 2\tau, \quad (10)$$

where K_T is the opto-electricity convert coefficient of detector and I_0 is the optic intensity accepted by detector.

Subtracting Eq. (9) from Eq. (7) yields

$$\Delta V_1 = V_1 - V_3$$

$$= -K_T I_0 \sin\left(\phi_R + \frac{\varepsilon\pi}{2}\right) - K_T I_0 \sin\left(\phi_R - \frac{\varepsilon\pi}{2}\right)$$

$$= -2K_T I_0 \sin(\phi_R) \cos\left(\frac{\varepsilon\pi}{2}\right). \quad (11)$$

Subtracting Eq. (10) from Eq. (8) yields:

$$\Delta V_2 = V_2 - V_4$$

$$= -K_T I_0 \cos(\phi_R + \varepsilon\pi) + K_T I_0 \cos(\phi_R - \varepsilon\pi)$$

$$= 2K_T I_0 \sin(\phi_R) \sin(\varepsilon\pi). \quad (12)$$

Then dividing Eq. (11) by Eq. (12) yields

$$\Delta X = \frac{\Delta V_1}{\Delta V_2} = \frac{2K_T I_0 \sin(\phi_R) \sin(\varepsilon\pi)}{-2K_T I_0 \sin(\phi_R) \cos(\varepsilon\pi/2)} = \frac{-\sin(\varepsilon\pi)}{\cos(\varepsilon\pi/2)}$$

$$= -2 \sin(\varepsilon\pi/2). \quad (13)$$

Therefore, the gain error in modulation chain is demodulated according to the detector signal in the four state modulation periods. Because the demodulation does not overlap with the feedback signal, the demodulation results have no relation with the serrodyne reset; that is, it will not be invalid at high and low angular rate in theory.

3.3 Realization of the Periodical Biasing Modulation Method

To demodulate and control the gain error in modulation chain by periodical biasing modulation method, a gain controlling circuit is appended to the usual controlling circuit of IFOG in Fig. 1. The appended controlling circuit of IFOG is shown in Fig. 4. Through the circuit in Fig. 4, the feedback serrodyne signal and periodical biasing signal are added alternatively in

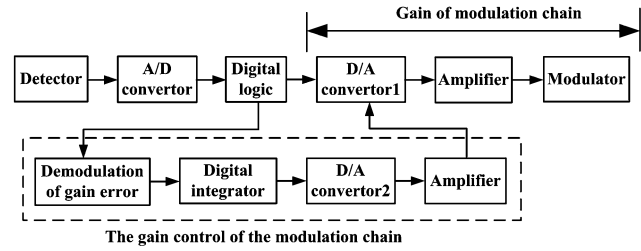


Fig. 4 The schematic of the gain control circuit.

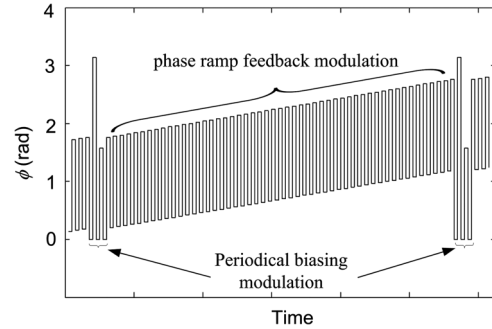


Fig. 5 The periodical biasing modulation signal and feedback modulation signal.

the controlling circuit of IFOG. The signal in the circuit is shown in Fig. 5. In the feedback serrodyne period, the biasing modulation signal is off, and the feedback signal is on so that the Sagnac phase shift is compensated, while in the biasing modulation period, the feedback signal is off, and the biasing modulation signal is on so that the gain error is demodulated. The demodulation results are feedback to D/A converter 1 to control its reference voltage after digital integral and D/A conversion.

4 Experiments

The temperature fluctuation is the main source of gain error in the modulation chain, so the validity of the demodulation method based on periodical biasing modulation is confirmed by temperature test. The IFOG with the controlling circuit

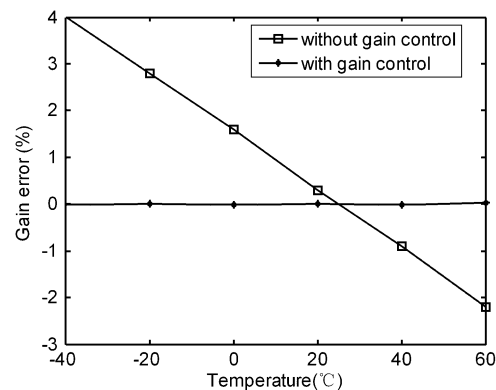


Fig. 6 The gain error with and without gain control at different temperature.

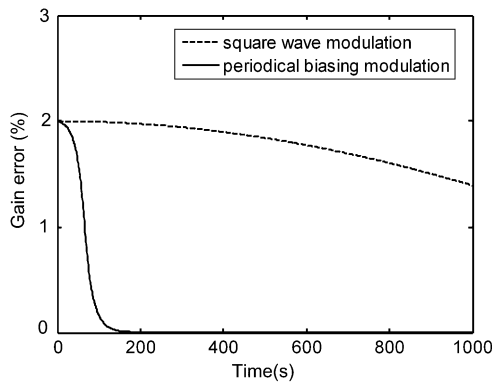


Fig. 7 The gain error controlled by square wave modulation and periodical biasing modulation method.

shown in Fig. 4 is put into a temperature control chamber, and the gain error in modulation is examined at different temperatures. The examined results are shown in Fig. 6. In the temperature range of -40 to about $+60^{\circ}\text{C}$, the variation of gain error is 6.2% . When the gain is not controlled, in contrast, the variation is 0.03% when the gain is controlled by periodical biasing modulation. Therefore, the gain error caused by temperature fluctuation can be suppressed effectively by the method.

To evaluate the validity of the method at low angular rate, the IFOG with modulation gain error 2% realized by tuning the gain of operational amplifier is fixed on a horizontal plane, the sensitive axis of which is oriented to east. The modulation gain error is demodulated by the periodical biasing modulation method and square wave modulation method respectively. The gain error in modulation chain is shown in Fig. 7. It is obvious that square wave demodulation method cannot recognize and control the gain error quickly so that the gain error cannot converge in controlling the circuit for a long time. However, the gain error tends to be zero fast when the periodical biasing modulation method is adopted, so this method is valid at low angular rate. To evaluate the validity of the method at high angular rate, the same IFOG oriented to be perpendicular to local horizon is fixed on a turn table, the angular rate of which is set to be 150 deg/s . The modulation gain error is demodulated by periodical biasing modulation and four-state wave modulation methods, respectively.

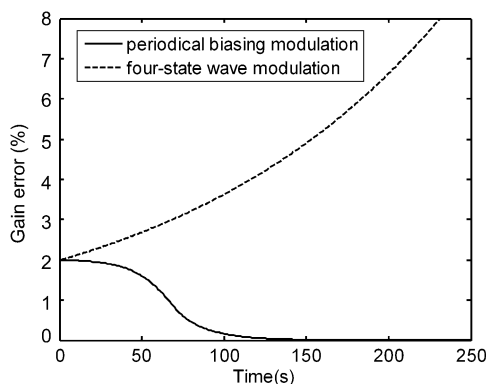


Fig. 8 The gain error controlled by four-state wave modulation and periodical biasing modulation method.

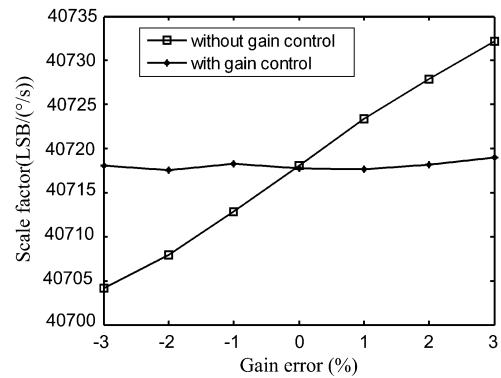


Fig. 9 The scale factor of FOG with and without gain control when the simulated gain error is added.

The gain error in modulation chain is shown in Fig. 8. From Fig. 8, the gain error is convergent when the periodical biasing modulation method is adopted but is not when four-state wave modulation method is adopted. Therefore, the periodical biasing modulation method also applies to high angular rate.

According to Eq. (1), the gain error influences the performances about scale factor of IFOG. The aim of controlling the gain error in the modulation chain is to improve the stability of scale factor under fluctuant temperatures. The scale factor depends on many parameters of IFOG; the direct test results of scale factor under fluctuant temperature cannot show distinctly what role the periodic biasing modulation plays. The benefit of the periodic biasing modulation method to scale factor is confirmed by the followed experiments. According to the above experiments, the gain error caused by temperatures fluctuating between -40 and approximately 6°C ranges from about 4% to -2% ; therefore, the gain error ranging from 4% to -2% is introduced in the modulation chain of IFOG by tuning operational amplifier to simulate the influence of temperature on the modulation gain. The scale factor and its nonlinearity of the IFOG with simulated temperature influence is tested, and the results are shown in Figs. 9 and 10. From both Figures, the stability of scale factor is decreased from 224 ppm to 7 ppm , and the nonlinearity of scale factor is decreased from 35 to 6 ppm due to the contribution of the periodic biasing modulation.

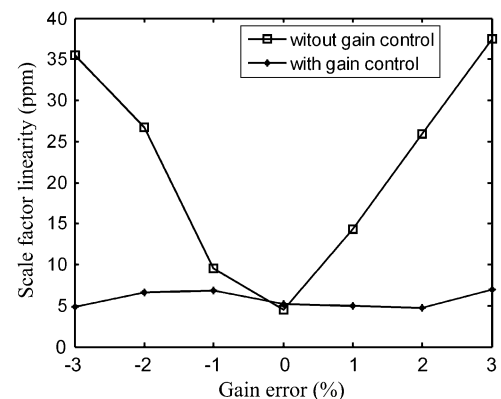


Fig. 10 The scale factor linearity of FOG with and without gain control when the simulated gain error is added.

5 Conclusion

The limitation of both existed methods to demodulate the gain error in the modulation chain of IFOG, the square wave modulation method and four-state wave modulation method, is revealed. The square wave modulation method will become invalid at low angular rate due to the long reset time and the four-state wave modulation method at high angular rate due to frequent reset. To compensate the defects of the existed methods, the periodic biasing modulation method is supposed and confirmed by experiment. The experimental results indicate that the supposed method can effectively control the gain error at low and high angular rates and benefit to the scale factor of IFOG.

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